NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
B.TECH 5TH SEM IT
AUTOMATA THEORY (ITPC37)
SESSIONAL-I

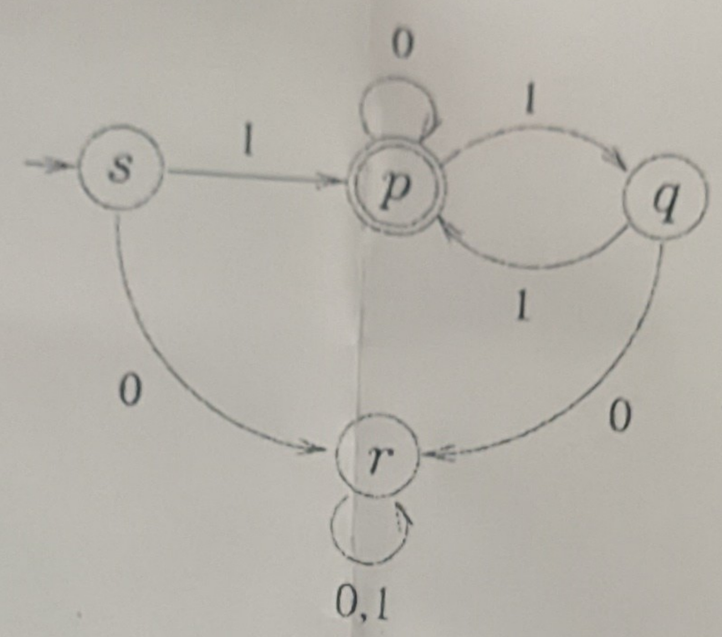
Duration: - 50 Minutes. **M.M 15**

Note: - All questions are compulsory.

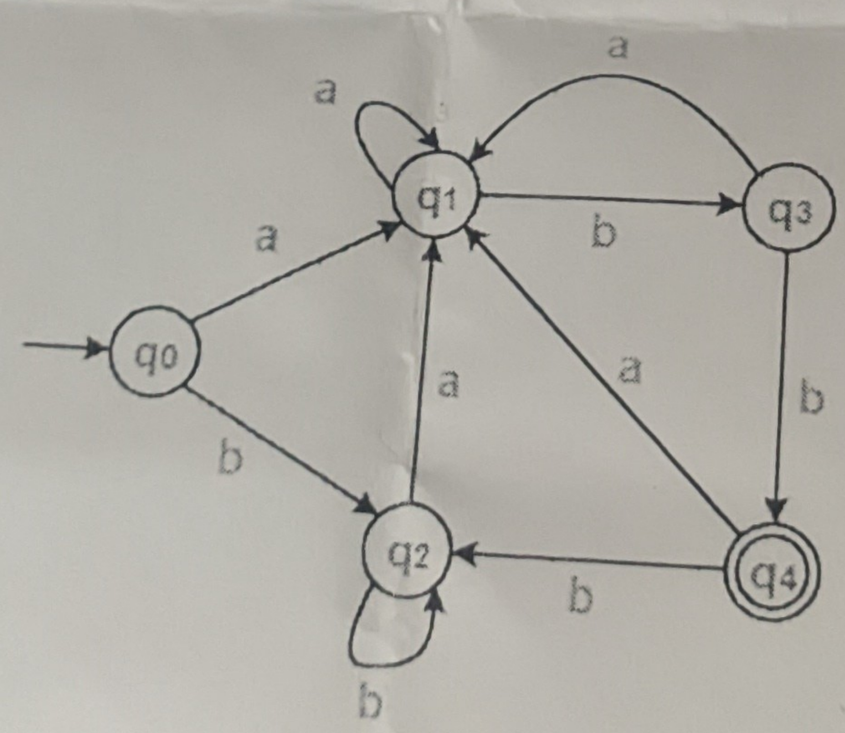
Q1 Draw the DFA for following languages: **(4)**

(i) L = { x ∈ {a,b}* number of a's in x divisible *by 2 but not divisible by 3.*
(ii) The language of all strings containing *both 11 and 010 as substrings.*

Q2 Convert following Finite Automata to Regular Expression using Arden's theorem. **(4)**



Q4 Minimize the no. of states in the following finite automata: **(4)**



Q Convert following NFA to DFA. Draw the final transition diagram **(3)**

Final state

δ	ε	a	b
→ q_0	$\{q_2\}$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
q_2	$\{q_0\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_2\}$